

DailyCare: An AI Homecare Agent for Elderly Heart Failure Management

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1 Introduction

Chronic Heart Failure (HF) is a condition that is especially prevalent among elderly patients, and it requires close monitoring of vital signs, symptoms and medication adherence. HF patients are also encouraged to adopt significant lifestyle changes such as restricting sodium and fluids, and participating in Exercise-based Cardiac Rehabilitation (ExCR) [1]. These interventions are especially challenging for elderly patients, who often forget to take their medications [2] or struggle to travel to the clinic regularly for checkups due to mobility issues. There is a need for real-time support and personalized advice for these patients as most patient management occurs outside the periodic doctor visits in clinical settings. DailyCare is an AI-powered homecare assistant designed to support heart failure patients through continuous monitoring, risk prediction and conversational guidance. The system allows users to log vital signs, track symptoms, and receive medication reminders and safe guidance for their health-related queries (through a conversational AI agent). It also analyzes trends in patient data to detect early warning signals of potential deterioration. By providing an integrated platform for these services, DailyCare aims to reduce uncertainty for HF patients and empower them to make more informed decisions.

GitHub Repository: <https://github.com/SahaDevShanmugam/DailyCare.git>

Demo Video: https://drive.google.com/file/d/1n-X6mmWasLtQqtxGSN0ARe_cLIspQJu/view?usp=sharing

2 System Architecture

2.1 User Interface

Figure 1 illustrates the overall system architecture of DailyCare. The elderly-friendly user dashboard (Figure 2) displays medication reminders, heart failure risk scores, current health status and customized, up-to-date recommendations. Users can enter their vital signs such as blood pressure, heart rate and temperature, and they can also track their weight, log symptoms, confirm medication intake and ask questions through the chat interface. The interface communicates with the backend through REST API calls.

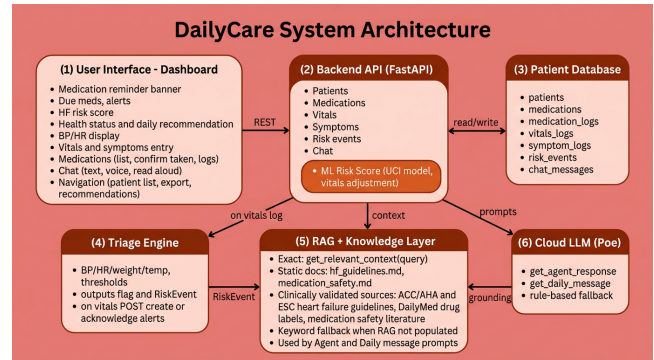


Figure 1. System architecture of DailyCare.

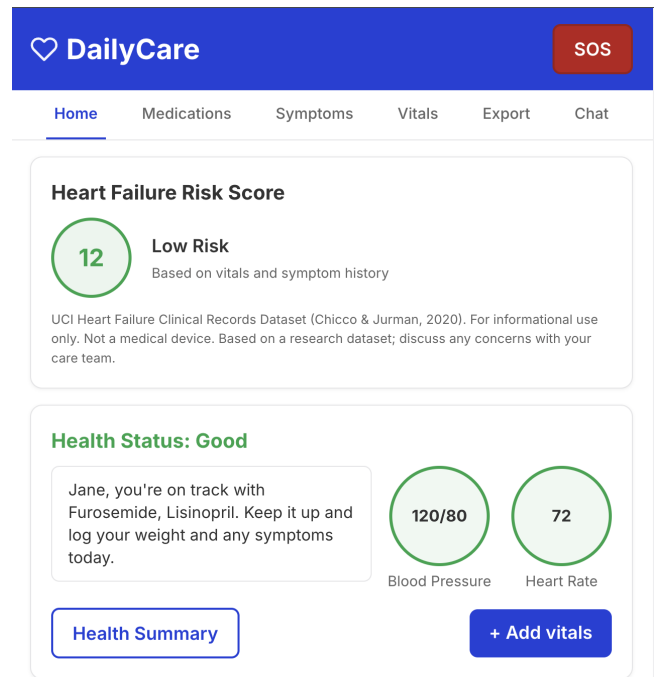


Figure 2. User Interface

2.2 Backend API

The backend API, implemented using FastAPI, acts as the central coordinator of the system. It receives inputs from the user interface, processes requests, and interacts with the patient database. The backend triggers risk prediction

updates when new vital signs are logged and forwards user queries to the agent pipeline.

2.3 Patient Database

The patient database continuously stores all the patient's health information required for monitoring and personalization. This includes patient records, medication lists, medication adherence logs, vital sign logs, symptom reports, detected risk events, and chat messages. The backend performs read and write operations to maintain up-to-date patient records.

2.4 Triage Engine

The triage engine evaluates all incoming health data for potential warning signs of HF deterioration. When new vitals are entered, the engine checks them against predefined thresholds for parameters such as blood pressure, heart rate, body weight and temperature. If abnormal patterns are detected, the engine generates a risk event which updates the patient's risk status and can trigger new alerts or recommendations.

2.5 RAG knowledge layer

The retrieval-augmented generation (RAG) layer provides grounded medical context for the conversational agent. When a user query is received, the system retrieves the relevant information from a curated medical knowledge base with heart failure management guidelines and medication safety references. These sources include the 2022 American Heart Association (AHA)/American College of Cardiology (ACC)/Heart Failure Society of America (HFSA) Guideline for the Management of Heart Failure [3], the European Society of Cardiology (ESC) heart failure guidelines [4], DailyMed Drug Labels [5], and PubMed [6]. The retrieved context is then passed to the language model to improve response reliability.

2.6 Large Language Model

DailyCare uses a cloud-hosted large-language model (LLM) accessed via Poe API, configured by default to use Claude Sonnet 4. For each request, the backend composes a rich prompt that combines a structured patient context (demographics and lifestyle, active medications with instructions and time of day, recent medication logs, vitals and symptom history, recent chat messages) with a knowledge context retrieved from the RAG layer. The model is prompted to personalize advice using this data, stay within the bounds of provided clinical knowledge, avoid diagnosis, and escalate to the care team when symptoms are concerning. This LLM powers both the interactive chat assistant and the short daily recommendation section on the main dashboard.

2.7 Machine Learning Risk Prediction

The heart-failure risk score is produced by a logistic regression model trained on the UCI Heart Failure Clinical Records

dataset [7] with standardization and median imputation, implemented in scikit-learn and persisted as `model.joblib` plus `feature_medians.json`. At inference time, the backend builds a feature vector for each patient using available app data (age, sex, recent systolic BP) and median defaults for unmeasured labs, then computes the probability of the UCI dataset's DEATH_EVENT outcome and scales it to a 0-100 risk score. This score is then categorized as low, moderate or high risk. The base probability is also adjusted using the latest vitals via the Triage Engine. Critical abnormalities (e.g. extremely high heart rate/BP or rapid weight gain) increase the score and force a higher tier, while less critical abnormalities force a moderate tier. This ensures that the displayed risk score reflects both the long-term model and the patient's most recent physiological status.

3 Key features

3.1 Medication reminders and adherence logging

The app helps patients stay on top of their HF medications by displaying a schedule with clear time-of-day labels, highlighting doses that are currently due, and requiring tap confirmation when a dose is taken. Each confirmation creates a medication log entry that is stored in the backend and reused by the assistant to comment on adherence patterns in the chat and within the recommendation section. The user also gets a notification at the appropriate time window for each medication (morning, afternoon, evening, night).

3.2 Vitals and Symptom tracking

Patients can enter daily vitals (blood pressure, heart rate, weight, temperature) and free-text symptoms using simple forms optimized for older adults. These entries are timestamped, stored in the patient database, and visualized on the dashboard as the latest blood pressure/heart rate rings and weight-change information, providing both the user and downstream algorithms with an up-to-date picture of their clinical status

3.3 Triage and alerting

Every new vitals entry is passed through the triage engine which applies configurable thresholds for blood pressure, heart rate, weight gain and temperature to classify the reading as "normal", "warning", or "critical". Critical values automatically generate structured risk events that surface on the dashboard as an alert card with plain-language explanations (e.g. rapid weight gain suggesting possible fluid retention).

3.4 Recommendations and health status summary

On the main dashboard, the "Health Status" panel summarizes whether the patient is stable or needs attention and provides a short, time-aware daily recommendation. This message is grounded in the latest vitals, symptoms, and medication logs plus guideline-based knowledge, and is phrased

as one or two concrete self-care actions (such as taking morning weight or limiting salt for the day).

3.5 Conversational assistant

The application includes an integrated assistant that patients can talk to either via text or speech input to ask questions about their medications, symptoms, or when to seek care. Behind the scenes, the assistant sees a consolidated view of the patient's history and knowledge-based snippets, allowing it to provide personalized, context-aware answers while reminding the user to contact their care team for urgent or uncertain issues. The responses can be read aloud for accessibility and can be stopped at any point during the playback or disabled entirely.

3.6 Data export for clinicians

For clinical follow-up and documentation, the system can export the patient's complete history (including vitals, symptoms, medications and adherence logs) as a ZIP file of CSVs (password-protected by the user). This export is intended for clinicians and/or researchers to quickly ingest into their own tools, review longitudinal patterns or include in their clinical records without requiring direct database access.

4 Agent Use Case Examples

4.1 Vital Monitoring Scenario

When a user enters elevated heart rate or sudden weight gain, the system flags the readings as high risk and updates the patient's risk score. It tells the user what they should do next (rest, check again, and contact the care team or emergency services if necessary).

4.2 Medication Question Scenario

A user asks whether two prescribed drugs (e.g. furosemide and desmopressin) can be taken together. Since the agent has access to a drug label database, it is aware of the possible side effects and contraindications of each drug. In this case, it was explicitly stated in the medication description that they should not be taken together and it correctly informs the user about this information.

4.3 Lifestyle Recommendation Scenario

A user asks for dietary guidance. The agent generates a full meal plan with low sodium levels and tells the user what kinds of food they should avoid.

5 Evaluation methods

5.1 Heart Failure Risk Score Accuracy

The risk score model is evaluated using standard classification accuracy on a held-out test set, ensuring that the model is tested on data it has never seen during training. The UCI Heart Failure dataset is first split into 80% for training and 20% for testing, using scikit-learn (train_test_split)

S. N	Items	Answer	Yes, N (%)	No, N (%)
1	Patients with HF should drink plenty of fluids each day	No	53(29.4)	127(70.6)
2	Processed lean meats are an acceptable food choice as part of the patient's diet	No	69(38.3)	111(61.7)
3	As long as no salt is added to foods there are no dietary restrictions for patient with HF	No	64(35.6)	116(64.4)
4	Coughing and nausea/poor appetite are common symptoms of advanced HF	Yes	137(76.1)	43(23.9)
5	Patient with HF should decrease activity and most forms of active exercise should be avoided	No	147(81.7)	33(18.3)
6	In case of weight gain of more than one kg during one day, this means HF become more worse	Yes	129(71.7)	51(28.3)
7	Swelling of the abdomen may indicate retention of excess fluid due to worsening of HF	Yes	147(81.7)	33(18.3)
8	If patients take their medication as directed and follow the suggested lifestyle modifications, their HF condition will not return	No	76(42.2)	104(57.8)
9	When a patient has aches and pain, aspirin and non-steroidal anti-inflammatory drugs (ibuprofen) should be recommended	No	105(58.3)	75(41.7)
10	If a patient adds extra pillows at night to relieve shortness of breath, this does not mean that their HF condition has worsened	No	65(36.7)	115(63.3)
11	If a patient wakes up at night with difficulty breathing, and the breathing difficulty is relieved by getting out of bed and moving around, this does not mean that the HF condition has worsened	No	61(33.9)	119(66.1)
12	Once the patient's HF symptoms are gone, there is no need for obtaining daily weight	No	70(38.9)	110(61.1)
13	When assessing weight results, today's weight should be compared with the patients' weight from yesterday, not the patients ideal or "dry" weight	No	83(46.1)	97(53.9)
14	Once heart failure signs disappear, daily weighing is no longer required	No	68(37.8)	112(62.2)
15	Eating large amount of fruits and vegetables, whole grains and low-fat dairy products is advised for patients with HF	Yes	145(80.6)	35(19.4)
16	Patients with HF are advised to take the diuretic medication at morning	Yes	143(79.4)	37(20.6)
17	BP recording of 80/56 without any HF Symptom should report	No	102(56.2)	78(43.3)
18	Dizziness or light headedness when arising that disappears within 5 min. should report	No	108(60)	72(40)
19	New onset or worsening of fatigue should report	Yes	153(85)	27(15)
20	New onset of worsening of leg weakness or decreased ability to exercise	Yes	154(85.6)	26(14.4)

Figure 3. Heart Failure Knowledge Questionnaire

with stratify=y and random_state = 42. Stratification ensures that the proportion of positive and negative cases (survival vs. death) is the same in both training and test sets, so the test set is representative of the overall dataset. The model (logistic regression) is trained only on the training portion after features are standardized with a StandardScaler. Once training is complete, the test set is passed through the same scaler and the trained model is used to generate predicted class labels with model.predict(X_test_s). Accuracy is then computed by comparing these predictions to the true labels in the test set and taking the fraction of cases the model got right. The model's test accuracy obtained using this method was 0.817.

5.2 Heart Failure Knowledge Questionnaire

Nurses working in a cardiac unit in Ethiopia were assessed using a knowledge questionnaire in 2020 [8]. As seen in the video demonstration, we picked 5 questions from the 20-item knowledge section of this test and posed them to the DailyCare AI agent from the user's perspective. Excerpts from the agent's replies are shown below.

Question 1: If my heart failure symptoms stop should I stop weighing myself? Answer: **"No, please don't stop weighing yourself even if your heart failure symptoms improve.** Daily weight monitoring is one of the most important tools for managing heart failure long-term, regardless of how you're feeling."

Question 2: Should I drink plenty of fluids everyday? Answer: **"Actually, with heart failure, "plenty of fluids" isn't always the best approach.** Since you're taking furosemide (a water pill) twice daily and I can see you're

currently experiencing moderate leg swelling, fluid management is really important for you. **Here's what I recommend:** - Follow any specific fluid limit your care team has given you (often 1.5-2 liters per day for heart failure patients) - If you haven't been given a specific limit, ask your doctor at your next visit - Spread your fluid intake throughout the day rather than drinking large amounts at once."

Question 3: Can I eat processed lean meat? Answer: **"I'd recommend limiting processed lean meats, even though they're lean.** Here's why this matters specifically for you: **With your current situation:** - You're taking furosemide twice daily and experiencing moderate leg swelling - You're managing heart failure, where sodium control is crucial - Your recent weight has fluctuated (77.0 kg to 75.0 kg) **The concern with processed meats:** - Even "lean" processed meats (like deli turkey, ham, or chicken) are typically very high in sodium - This extra sodium can worsen fluid retention and make your leg swelling worse - It can also make your furosemide less effective at removing excess fluid."

Question 4: Is coughing and nausea a symptom of heart failure? Answer: **"Yes, both coughing and nausea can be symptoms of heart failure,** but they can also be side effects of your medications, so it's important to figure out which is which."

Question 5: How much weight gain per day indicates that heart failure is getting worse? Answer: **"For heart failure patients like you, a weight gain of 2-3 pounds (about 1-1.4 kg) in one day, or 5 pounds (about 2.3 kg) in a week is concerning** and suggests your heart failure may be worsening."

The DailyCare AI agent was able to answer all the questions correctly and provide reasonable explanations and further suggestions to the user. It was also able to take the user's specific circumstances into account (e.g. "Your recent weight has fluctuated") by accessing their previous entries.

6 Conclusion

DailyCare demonstrates that combining structured patient data, clinical triage, machine learning risk estimation and a cloud LLM-based agent can produce a cohesive home-monitoring experience for people living with chronic heart failure. While it is not intended to replace clinical judgment, the prototype shows how guideline-grounded generative AI and traditional ML can be layered on top of routine self-monitoring to help patients understand when things are stable, when to be more cautious, and when to reach out to their care team.

7 Limitations and Future Improvements

At the same time, the current implementation has important limitations and clear avenues for future work. The risk model relies on a single public dataset and imputes several

lab values from medians, so its calibration and applicability to diverse populations remains uncertain. Extending it with real-world data, evaluating additional metrics (e.g. AUC, calibration plots) and exploring more expressive yet interpretable models would strengthen its clinical relevance. The RAG layer currently covers only a narrow set of heart failure and medication safety documents, and even with careful prompting the cloud LLM may still oversimplify or hallucinate, suggesting a need for a broader, regularly updated knowledge base and stricter guardrails for high-risk advice. Future iterations should focus on deployment-ready security, EHR integration, and prospective studies to determine whether DailyCare actually improves adherence, symptom recognition and downstream clinical outcomes.

8 References

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